

German University in Cairo

Mechatronics Lab (MCTR704)

Pick and Place

Project No. [53]

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Project Description

Project Idea:

The idea of this project is to design and build a Pick and Place Mechanism that uses pneumatic systems to automate the movement of objects from one location (Position X) to another (Position Y) that is at a higher elevation. The system incorporates three pneumatic cylinders, each responsible for specific movements.

Objective:

The primary objective is to develop a system capable of:

1. Picking objects from a specified location (Position X).
2. Lifting the objects vertically to a higher position.
3. Placing the objects at a new location (Position Y) using automated movements controlled by pneumatic cylinders.

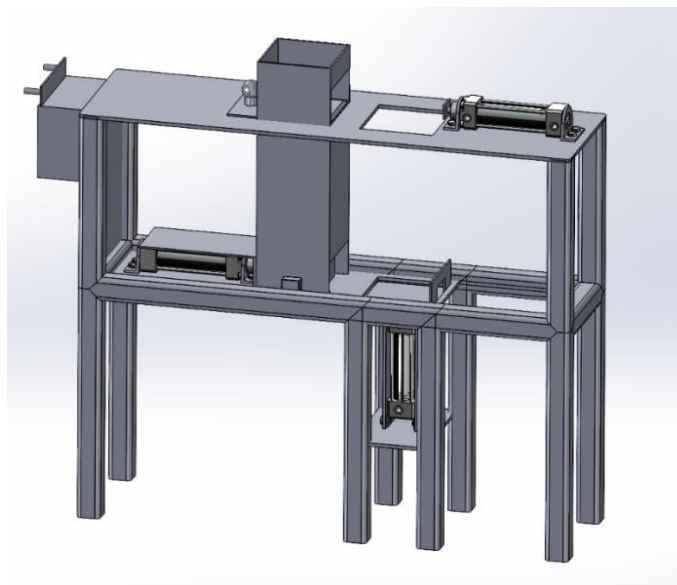
The system will integrate sensors.

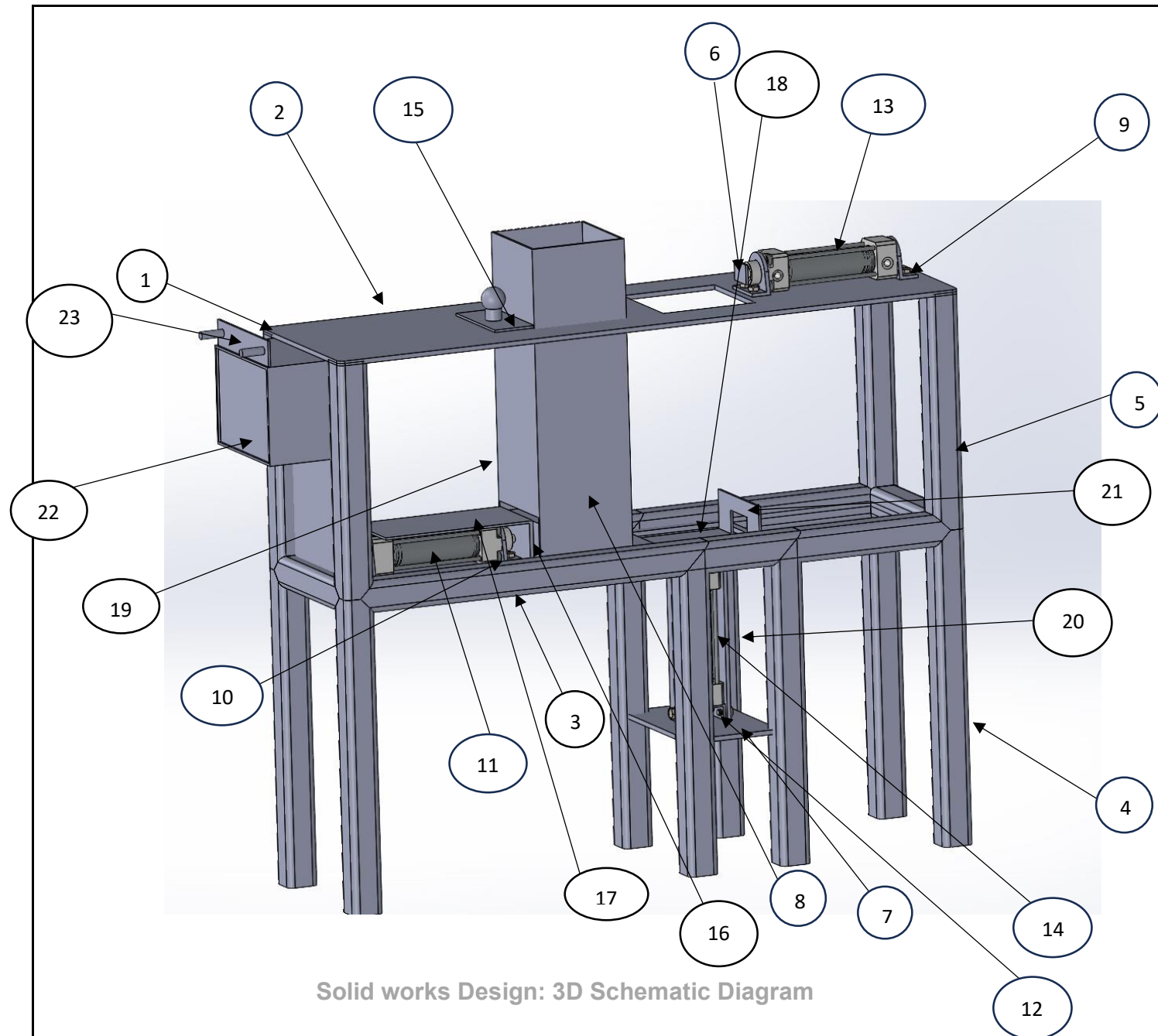
How it works:

1. Cylinder A dispenses objects from the magazine individually to point X.
2. Cylinder B, a vertical piston cylinder to push and elevate the object from point X to the desired high.
3. Cylinder C, a horizontal cylinder, is tasked with transporting the object to the designated location at point Y.

Project size:

Our project will be 100 x 40 x 80 cm

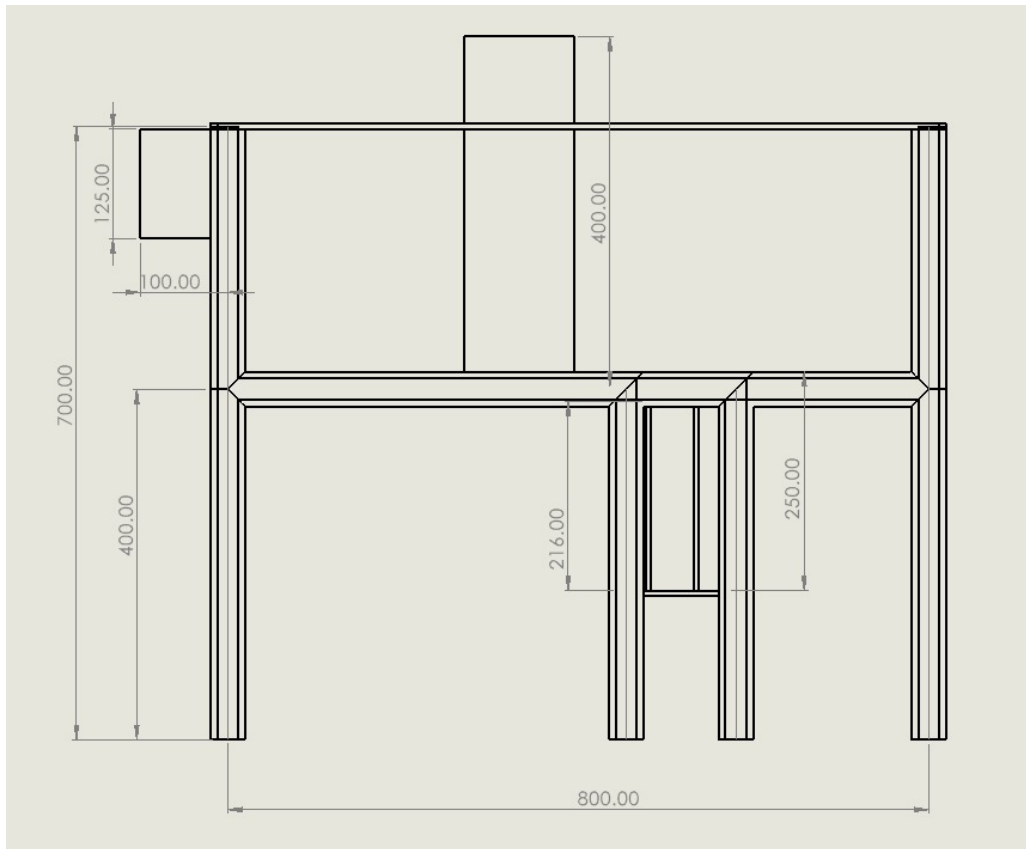


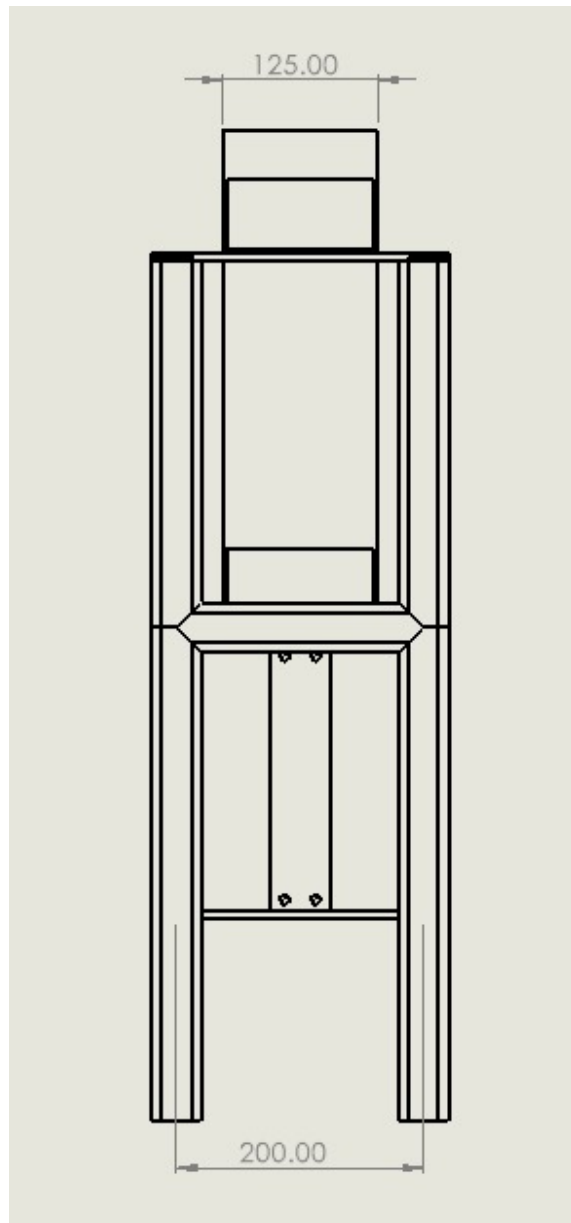


Part number	Name
1	Structure level two – 200 mm
2	Structure level two – 800 mm
3	Structure level one – 800 mm
4	Structure support bar – 40 cm
5	Structure support bar – 30 cm
6	Cylinder C end effector
7	Support to hold cylinder B
8	Stack for the objects – 125x125 mm
9	Cylinder C fixation

10	Cylinder A fixation
11	Cylinder A (length:192.7mm / Max extension: 139.77mm / Cross-sectional area: 47x47mm)
12	Cylinder B fixation
13	Cylinder C (length:192.7mm / Max extension: 139.77mm / Cross-sectional area: 47x47mm)
14	Cylinder B (length:192.7mm / Max extension: 139.77mm / Cross-sectional area: 47x47mm)
15	A port
16	Object (50x50x50 mm)
17	Cylinder A end effector
18	Moving table (cylinder B end effector)
19	Sensor fixation (in the back)
20	Support for cylinder B
21	Box stopper
22	Circuit box
23	Circuit box port

Mechanical Components 2D Projections with Dimensions



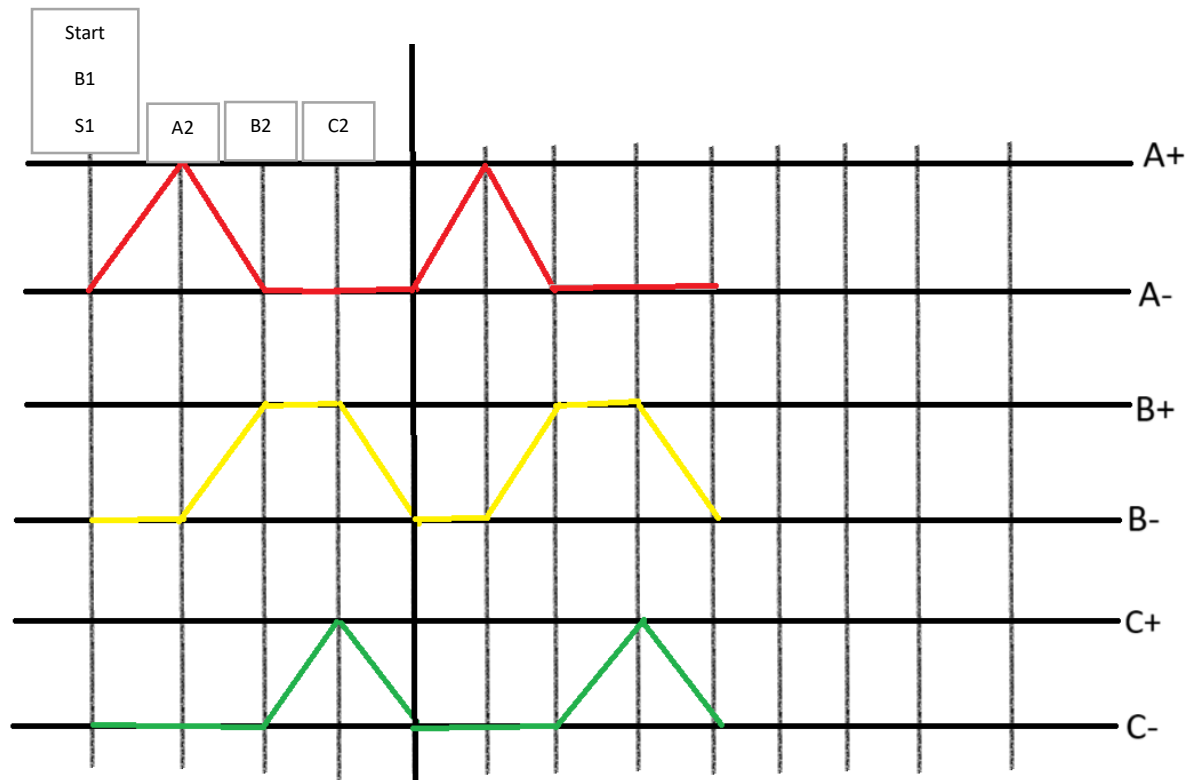


Project Components List and Data Sheets Description

	Name	Description	Quantity	Photo
1	Pneumatic cylinder (length:192.7mm / Max extension: 139.77mm / Cross-sectional area: 47x47mm)	Provide motion, whether for pushing or lifting	3	
2	Sensor (photocell sensor)	To detect objects (metallic and non-metallic)	1	
3	Relay	Controls high-power circuits using a low-power signal	7	
4	On and Off switch	To switch the on and off of the operation	1	
5	Valves (5/2)	To put them before the cylinders	3	
6	Reed switch	Play the role of a sensor to detect the cylinders when they are fully extracted or retracted	6	

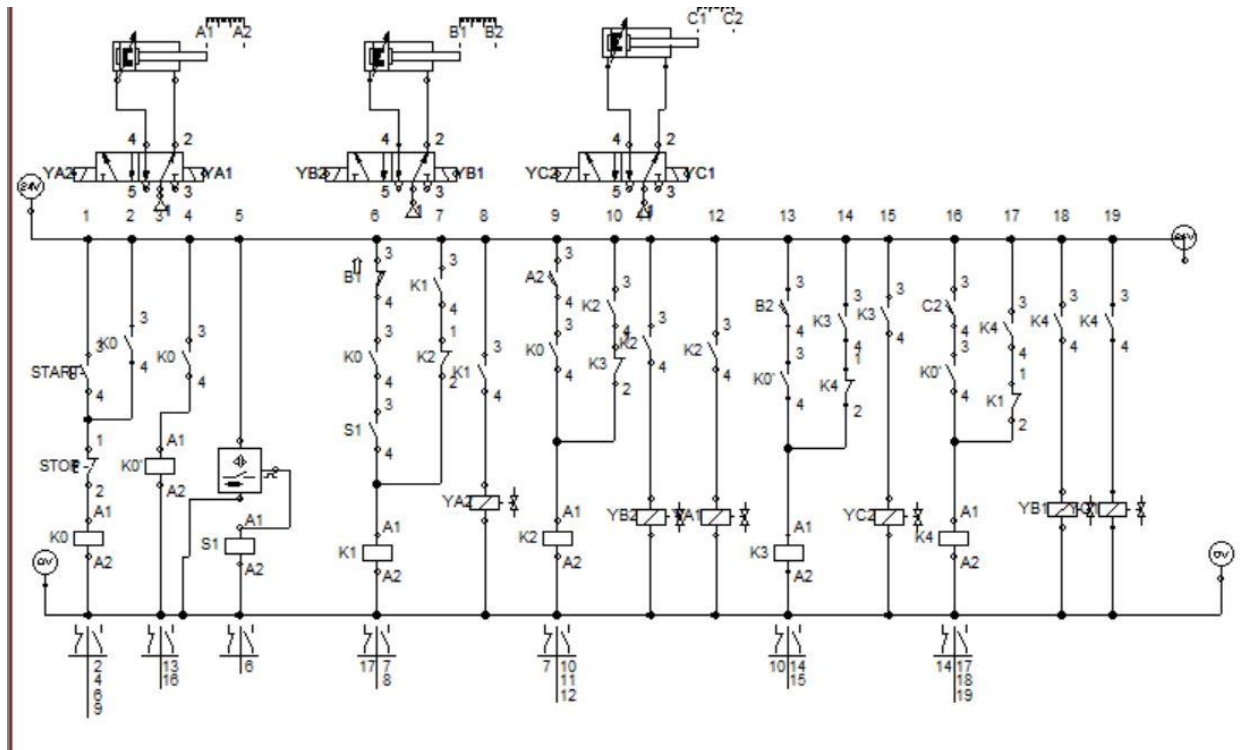
Pneumatic Step Diagram and Description

A+ (B+ A-) C+ (B-C-)



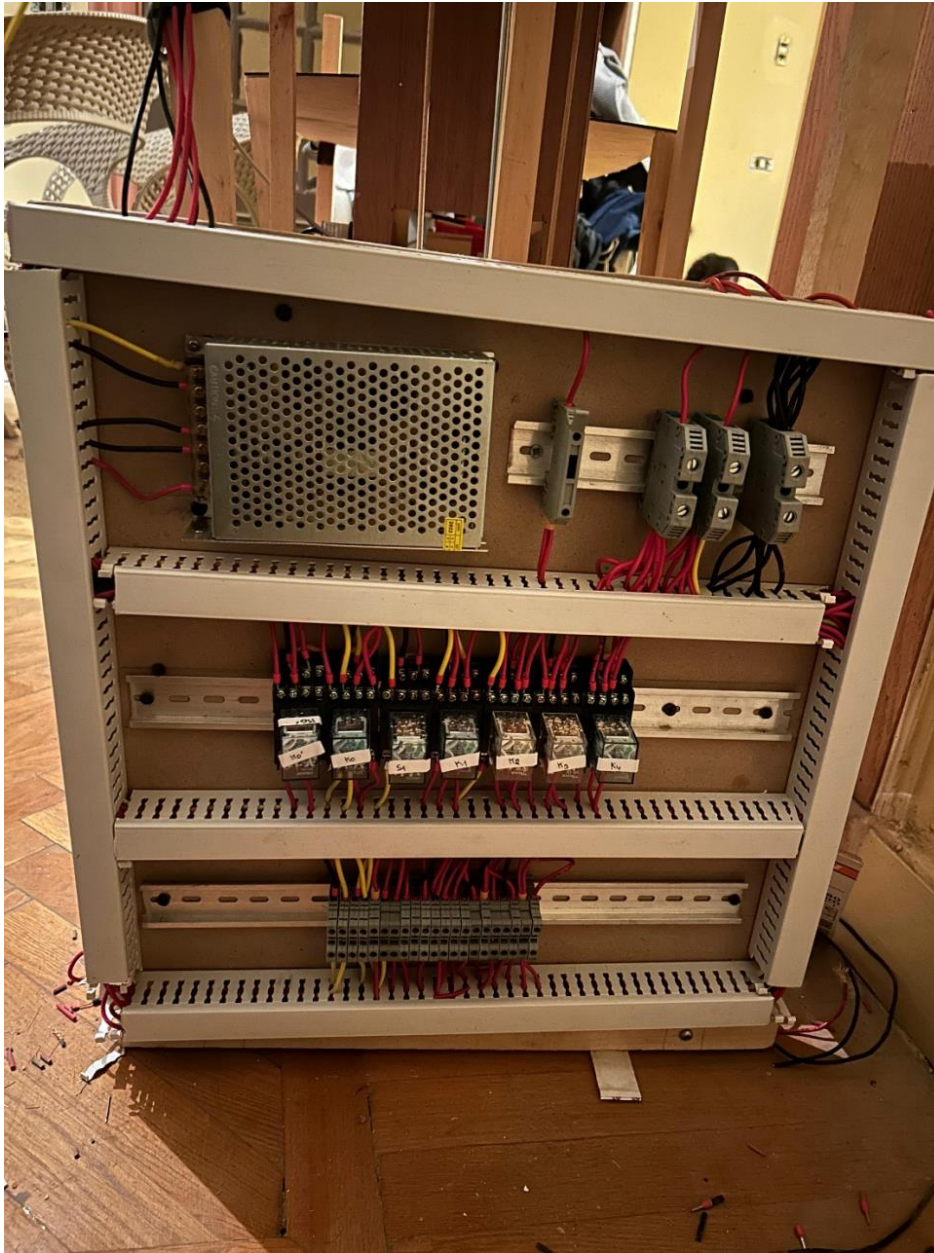
- When the **start** push button is pressed and the output of the **sensor** is 1 (the sensor detects that there is an object) and cylinder B is fully retracted (**B1**) → Cylinder A will extend (to push the object until it stops on the movable table).
- When cylinder A is fully extended (**A2**) → cylinder B will extend (to push the movable table with the object up) and cylinder A will retract at the same time.
- When cylinder B is fully extended (**B2**) → Cylinder C will extend (to place the object at the right position).
- When cylinder C is fully extended (**C2**) → Cylinder B and Cylinder C will retract together.

Electro Pneumatic Circuit



Hardware Model (Electrical + Mechanical)

Controller Operating Panel/ Classic Control Implementation



Control panel: 50 cm x 50cm

the list of materials

- Wood
- L bracket
- Screws
- Fixations
- Balsa wood

Project Documentation

Mechatronics Lab MCTR704

Project Hardware As Built

